

Sam Obila Allela

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PROJECT EXPERIENCE

Personal Projects

Data Scientist / AI Engineer

- Developed an end-to-end data analysis pipeline in Python using Pandas, Matplotlib, and Jupyter, covering data ingestion, cleaning, exploratory analysis, statistical summarization, and visual insight generation on real-world demographic datasets.
- Built a supervised ML classification model for sign language detection using Scikit-Learn and NumPy, including feature engineering, cross-validation, and performance benchmarking.
- Created a regression model to predict sea level outcomes from historical data, utilizing Scikit-Learn and NumPy, with hyperparameter tuning and residual analysis.
- Designed interactive data visualizations for medical datasets using Seaborn, Matplotlib, and Pandas to uncover distributions, outliers, and correlations.
- Engineered Study Bot, an AI-powered learning assistant leveraging Google ADK, Python, LLMs, and agent frameworks to deliver dynamic Q&A, concept explanations, and adaptive quizzing for personalized learning sessions.

EDUCATION

Self-Directed & Formal Study

Mathematics & Computer Science

LEADERSHIP EXPERIENCE

Build with AI Nairobi

Nairobi, Kenya

Speaker & Active Contributor

- Engaged the local AI community through talks and workshops, simplifying complex ML/AI concepts.
- Multiplied technical impact through peer education and community outreach.

SKILLS & INTERESTS

Skills: Data Science & Analytics (EDA, statistical analysis, hypothesis testing, predictive modeling, data wrangling); Machine Learning (classification, regression, clustering, feature engineering, cross-validation, ML pipelines); AI Engineering (LLM integration, RAG pipelines, Pinecone vector DB, LangChain, prompt engineering); Mathematics & CS (probability, linear algebra, statistics, algorithms, computational thinking); Education & Communication (technical simplification, community workshops, public speaking); Technical: Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Jupyter, LangChain, Pinecone, RAG Pipelines, Hugging Face, Google ADK, TensorFlow, PyTorch, Plotly, Git, GitHub, VS Code.

Interests: AI community engagement, technical education, public speaking, continuous learning in deep learning frameworks (TensorFlow, PyTorch), data visualization (Plotly, Seaborn), and open-source contributions.